Republic of the Philippines
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Automated Garage Door using Ultrasonic Sensor

In Partial Fulfillment of the Requirements for Applications Development & Emerging Technologies

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THE PROBLEM AND ITS BACKGROUND

Introduction

In modern times, automation has significantly contributed to enhancing convenience and security in various applications. Automatic doors are widely used in residential, commercial, and industrial spaces due to their ability to provide hands-free operation, especially in high-traffic areas. This project explores the development of an automated garage door system powered by an ultrasonic sensor, a button for manual operation, and an integrated buzzer for user alerts. The goal of this project is to design an efficient, user-friendly, and reliable system that responds to proximity and manual inputs, ensuring smooth operation while enhancing accessibility and safety.

Statement of the Problem

The traditional manually operated doors often pose challenges in terms of safety, accessibility, and convenience, particularly in high-traffic or hands-busy environments. These challenges can lead to inefficiencies, reduced comfort, and safety concerns for users, such as difficulty in handling doors while carrying objects or individuals with physical disabilities.

This project addresses the following key problems:

1. Limited Accessibility: How can a door system be designed to automatically operate without requiring physical contact, thus improving accessibility for users, including those with mobility impairments?
2. Safety Concerns: How can the garage door system ensure it remains open as long as it detects an object or person underneath, reducing the risk of accident.
3. Enhanced Awareness: How can the system notify users when the door opens and closes to improve awareness and security?
4. Integration of Manual Control: How can manual operation via a button be effectively integrated for cases where automatic operation is not preferred or feasible?

Objectives of the Study

The primary objective of this project is to design and develop a prototype of an automated garage door system that enhances safety, accessibility, and convenience for all users, including individuals with disabilities.

Specifically, the project aims to:

1. To implement an ultrasonic sensor-based detection system that ensures the door remains open as long as an object or person is detected beneath it.
2. To enhance safety by preventing the garage door from closing prematurely when an object or person is detected.
3. To add additional features, such as a buzzer and button, to provide manual control and audible notifications for users.

Scope and Limitation

This project focuses on the design and development of an automated garage door system aimed at improving accessibility, safety, and convenience for users. The system integrates an ultrasonic sensor to detect objects or people beneath the door, ensuring it stays open when necessary. Additional features include a manual control button and a buzzer for audio alerts when the door operates. The design is tailored for residential use and prioritizes users with mobility challenges, such as individuals with disabilities or those managing heavy loads, emphasizing ease of operation and safety.

However, the system is a prototype and may not fully replicate a large-scale garage door's mechanical and structural requirements. Environmental factors such as extreme weather conditions or sensor interference (e.g., by reflective surfaces or low-light conditions) might affect the ultrasonic sensor’s accuracy. The manual button is designed for simple operation, but it might lack advanced features such as remote or smartphone integration. Furthermore, the system assumes controlled indoor or semi-outdoor environments and may not account for tampering or unauthorized access.

Review of Related Literatures

Book

Ultrasonic sensors operate by emitting ultrasonic pings and listening for the sound waves that bounce back from objects. This principle mimics a bat's sonar system. While effective, this technique has limitations. Certain textures or shapes may not reflect sound waves properly—soft surfaces like pillows may absorb the waves, while angled objects might deflect them. Different models of ultrasonic sensors, such as the Parallax Ping or MaxBotix series, vary in range, sensitivity, and detection angles. For instance, the Parallax Ping has a range of up to 3 meters with a narrow detection angle, whereas the MaxBotix EZ-0 provides a broader sensing area, suitable for monitoring wide zones. Users are advised to consult a sensor's specifications to select one suited for their application. Additionally, environmental factors like temperature can affect the accuracy of distance measurements, highlighting the need for careful calibration. (Baichtal, 2014, p. 259-260)

URL

Automatic doors were first introduced in 1931 by engineers George C. Herrick and Charles E. Brown, using pneumatic sensors for motion detection. Over time, the technology has advanced with radar and infrared sensors, making automatic doors common in public spaces. They offer benefits such as improved safety, hygiene, energy efficiency, and accessibility, especially for people with mobility challenges. These features align with your project, where your automatic garage door system enhances safety and convenience, similar to the broader trend of improving accessibility through automated door systems. (Gaulton, 2024)

The article, " A Smart Home Appliance Control System for Physically Disabled People" (Mtshali et al., 2019) discusses the rise of "smart homes" facilitated by advancements in technologies initially designed to enhance convenience for non-disabled individuals. Smart homes have proven effective for improving security, energy efficiency, and convenience by enabling residents to control devices such as lighting, HVAC systems, and door locks remotely. However, most devices are not optimized for people with disabilities or limited mobility. For individuals with physical disabilities or the elderly, the ability to control home devices with minimal physical effort could significantly improve quality of life. The proposed system incorporates smart plugs, smart cameras, smart power strips, and digital assistants to allow users

with physical disabilities to control electrical appliances through sensors, offering an inclusive solution with minimal physical exertion.

Journal

The study "Design and Development of an Automatic Door Gate Based on Internet of Things Using Arduino Uno" (Hasibuan et al., 2021) focuses on creating an automated gate system leveraging IoT technologies. The system integrates an Arduino Uno, sensors, and a servo motor to detect motion and operate the gate. The goal is to enhance convenience and security for residential or commercial settings. Key features include motion detection using sensors to open and close the gate automatically, with a predefined delay for secure operation. The system also supports integration with IoT for remote monitoring and control, ensuring accessibility and modern functionality.

The article “Smart automatic COVID door opening system with contactless temperature sensing” (Venkataramanan et al., 2023) discusses the design of a prototype automatic door control system, aimed at reducing physical contact with door handles to prevent the spread of COVID-19. The system combines various sensors and embedded equipment to automate door opening based on human movement, body temperature, and the presence of a mask. The system aims to enhance safety and hygiene by minimizing physical contact with door handles, especially in public or healthcare environments.

Design Phase

Schematic Diagram

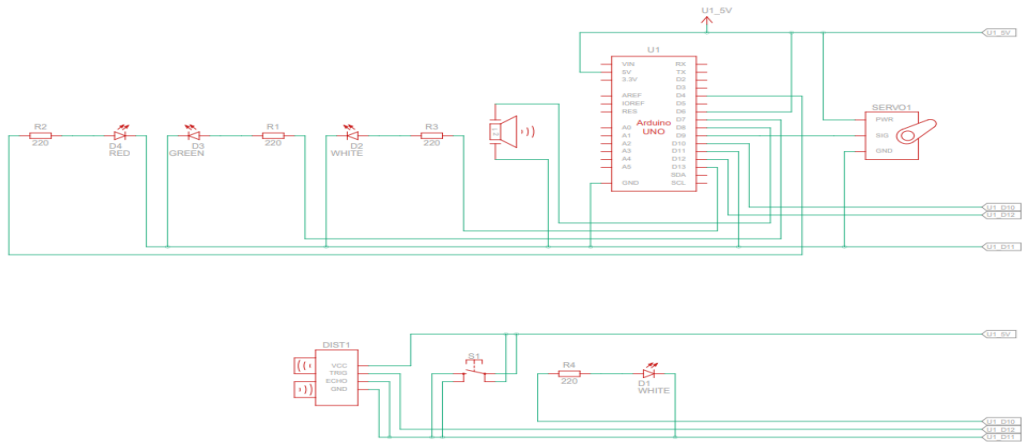


Figure 1: Schematic Diagram of the Automated Garage Door

The schematic diagram (shown in Figure 1) provides a detailed representation of the electrical connections within the automatic garage door system. It shows how the ultrasonic sensor, servo motor, LEDs, and other components are connected to the Arduino board for proper functionality.

Circuit Diagram

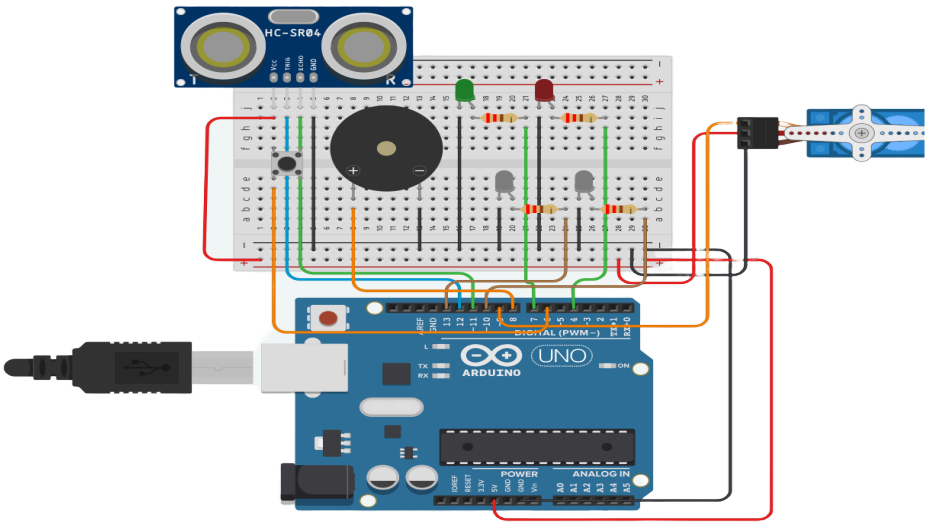


Figure 2: Circuit Diagram of the Automated Garage Door

The circuit diagram (shown in Figure 2) illustrates the connections and components of the automatic garage door system, including the ultrasonic sensor, servo motor, LEDs, buzzer, button, and Arduino board. It highlights how these elements are integrated to detect objects, control door movement, and provide visual and auditory indicators.

Development Phase

Initial Prototype

The first prototype focuses on using the ultrasonic sensor to detect objects and trigger the positional movement of the servo motor based on the measured distance. With the additional feature of 2 LEDs that will indicate if the door is opening (green LED) or closing (red LED).

```
#include <Servo.h>
#define trigPin 12
#define echoPin 11
#define openLed 7
#define closeLed 4
Servo servo;
int sound = 250;
int pos = 0;
void setup() {
    Serial.begin (9600);
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    pinMode(openLed, OUTPUT);
    pinMode(closeLed, OUTPUT);
    servo.attach(9);
}
void loop() {
    long duration, distance;
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);
```

```
    duration = pulseIn(echoPin, HIGH);
    distance = (duration/2) / 29.1;

  if (distance < 10) {
    digitalWrite(openLed, HIGH);
    digitalWrite(closeLed, LOW);

    for(pos=0;pos<=180;pos++){
      servo.write(pos);
      delay(20);
    }
  Serial.println("the distance is less than 10");
  delay(1500);

    for(pos=180;pos>=0;pos--){
      digitalWrite(openLed, LOW);
      digitalWrite(closeLed, HIGH);
      servo.write(pos);
      delay(20);
    }
  }
  else {
    servo.write(0);
  }
  if (distance > 60 || distance <= 0){
    digitalWrite(openLed, LOW);
    digitalWrite(closeLed, LOW);
    Serial.println("The distance is more than 60");
  }
  else {
  Serial.print(distance);
  Serial.println(" cm");
  }
  delay(500);
}
```

Second Prototype

The second prototype enhances the functionality by ensuring the garage door remains open as long as the ultrasonic sensor detects an object beneath it, adding a safety feature to prevent accidental closures.

```
#include <Servo.h>
#define trigPin 12
#define echoPin 11
#define openLed 7
#define closeLed 4
Servo servo;
int pos = 0;
bool doorOpen = false;

void setup() {
    Serial.begin (9600);
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    pinMode(openLed, OUTPUT);
    pinMode(closeLed, OUTPUT);
    servo.attach(9);
    servo.write(0);
}
void loop() {
    long duration, distance;
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    duration = pulseIn(echoPin, HIGH);
    distance = (duration/2) / 29.1;

  if (distance < 8 && distance > 0 && !doorOpen) {
        digitalWrite(openLed, HIGH);
        digitalWrite(closeLed, LOW);

        for (pos = 0; pos <= 180; pos++) {
            servo.write(pos);
            delay(20);
        }
        doorOpen = true;
        Serial.println("Door is open");
    }
```

```
    else if (distance > 9 && doorOpen) {
        digitalWrite(openLed, LOW);
        digitalWrite(closeLed, HIGH);

        for (pos = 180; pos >= 0; pos--) {
            servo.write(pos);
            delay(20);
        }
        doorOpen = false;
        Serial.println("Door is closed");
    }
    if (distance > 9 || distance <= 0){
      digitalWrite(openLed, LOW);
      digitalWrite(closeLed, LOW);
      Serial.print(distance);
      Serial.println(" cm");
  }

  delay(500);
}
```

Final Prototype

In the final prototype, additional features are added. A button for manual control and a buzzer for audible notification. The distance is also now set for the actual model of the project.

```
#include <Servo.h>
#define trigPin 12
#define echoPin 11
#define openLed 7
#define insideLed 13
#define insideLed2 10
#define closeLed 4
#define buttonPin 6
#define buzzerPin 8
Servo servo;
int pos = 0;
bool doorOpen = false;
```

```
void setup() {
    Serial.begin (9600);
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    pinMode(openLed, OUTPUT);
    pinMode(closeLed, OUTPUT);
    pinMode(buttonPin, INPUT_PULLUP);
    pinMode(buzzerPin, OUTPUT);
    servo.attach(9);
    servo.write(0);
}
void loop() {
    long duration, distance;
    bool buttonPressed = !digitalRead(buttonPin);

    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

     duration = pulseIn(echoPin, HIGH);
    distance = (duration/2) / 29.1;

  if (distance < 8 && distance > 0 && !doorOpen) {
        digitalWrite(openLed, HIGH);
        digitalWrite(insideLed, HIGH);
        digitalWrite(insideLed2, HIGH);
        digitalWrite(closeLed, LOW);
        activateBuzzer();

        for (pos = 0; pos <= 140; pos++) {
            servo.write(pos);
            delay(15);
        }
        doorOpen = true;
        Serial.println("Door is open");
        Serial.print(distance);
        Serial.println(" cm");
    }
    else if (distance > 8 && doorOpen) {
```

```
        digitalWrite(openLed, LOW);
        digitalWrite(insideLed, LOW);
        digitalWrite(insideLed2, LOW);
        digitalWrite(closeLed, HIGH);

        for (pos = 140; pos >= 0; pos--) {
            servo.write(pos);
            delay(30);
        }
        doorOpen = false;
        Serial.println("Door is closed");
    }
    if (distance > 8 || distance <= 0){
      digitalWrite(openLed, LOW);
      digitalWrite(closeLed, LOW);
      Serial.print(distance);
      Serial.println(" cm");
  }

      delay(500);
      if(buttonPressed) {
      if(!doorOpen){
        digitalWrite(openLed, HIGH);
        digitalWrite(insideLed, HIGH);
        digitalWrite(insideLed2, HIGH);
        digitalWrite(closeLed, LOW);
        activateBuzzer();

        for (pos = 0; pos <= 140; pos++) {
            servo.write(pos);
            delay(25);
        }
      }
      doorOpen = true;
      }
}
void activateBuzzer(){
  digitalWrite(buzzerPin, HIGH);
  delay(1000);
  digitalWrite(buzzerPin, LOW);
}
```

Testing Phase

The testing phase of the automated garage door system validated its functionality under various scenarios, including object detection, manual operation, and safety mechanisms.

Figure 3: Object Detection and Door Operation.

When an object, such as a vehicle or person, was detected beneath the ultrasonic sensor (shown in figure 3):

- The green LED illuminated, indicating the detection of the object.
- The buzzer emitted a short beep to notify users of the door's opening.
- The servo motor rotated to open the door fully, ensuring smooth accessibility.

If the object remained within the detection range of the ultrasonic sensor, the door stayed open, preventing premature closure and ensuring safety.

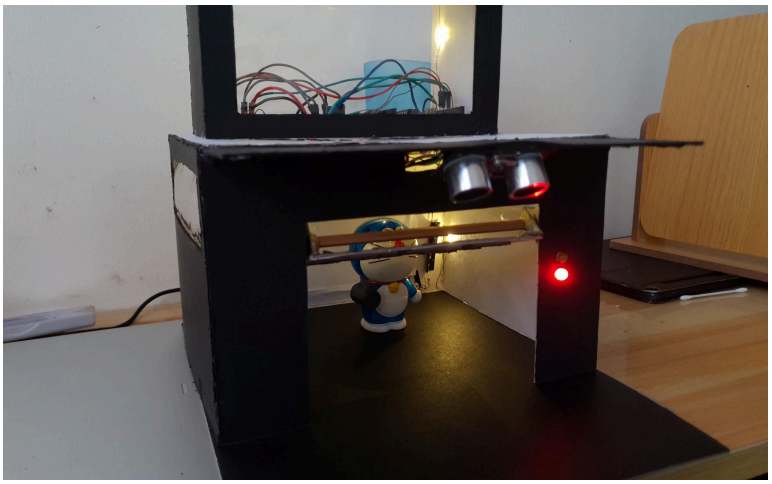
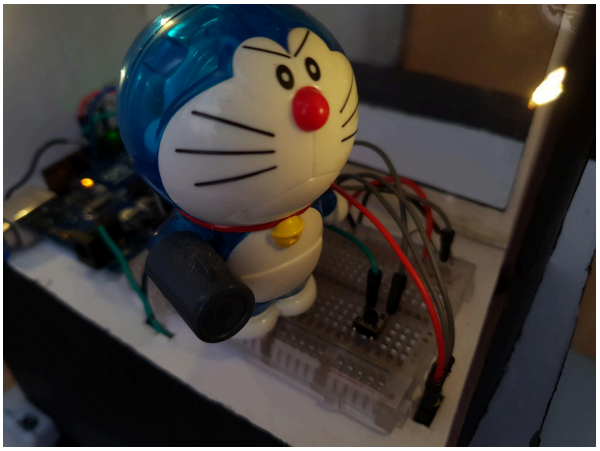


Figure 4: Object Moves Inside.

Once the detected object moved out of the ultrasonic sensor's range (shown in figure 4):

- The red LED illuminated, signaling readiness for the door to close.
- The servo motor rotated to close the door securely after verifying that no objects remained within the detection zone.



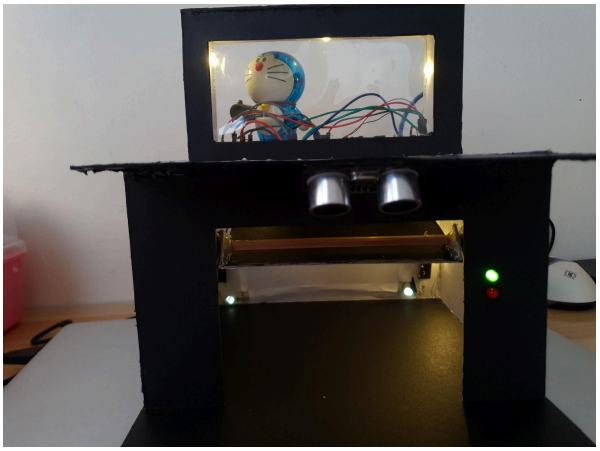


Figure 5 and Figure 6: Manual Operation.

A manual override feature was tested using the integrated button (shown in figure 5 and figure 6):

- Pressing the button allowed users to open or close the door, independent of the automatic mode.
- LED indicators (green or red) displayed the current door state.
- The buzzer emitted a notification during each operation to enhance user awareness.

This testing phase confirmed that the system met its objectives for safety, accessibility, and user control, ensuring reliable operation under automatic and manual modes.

Results

The survey gathered feedback from 10 respondents, all students from Our Lady of Fatima University-Laguna, regarding the automated garage door system. The purpose of the survey was to evaluate the system's usability, safety, accessibility, and overall effectiveness. Respondents were asked about their experiences with the system's ease of use, how safe they felt while operating it, and whether it improved accessibility for individuals with mobility challenges. The collected data offers valuable insights into the system's performance, highlighting both its strengths and areas for potential improvement.

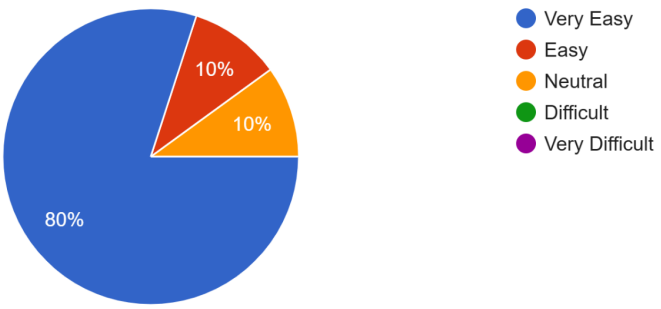


Figure 7: User Feedback on Ease of Use

The survey results (shown in figure 7) demonstrate that the automated garage door system is user-friendly. When asked, *"How easy was it to understand and use the automated garage door system?"* 80% of respondents indicated that it was "Very Easy" to understand and operate. Additionally, 10% rated the system as "Easy," reflecting the accessibility of its design. However, 10% selected a neutral response, suggesting that while the system is largely intuitive, minor enhancements could further improve usability. Overall, 90% of respondents found the system to be easy to use, underscoring its effectiveness in achieving one of the project's primary objectives: ensuring accessibility and ease of use for a diverse range of users.

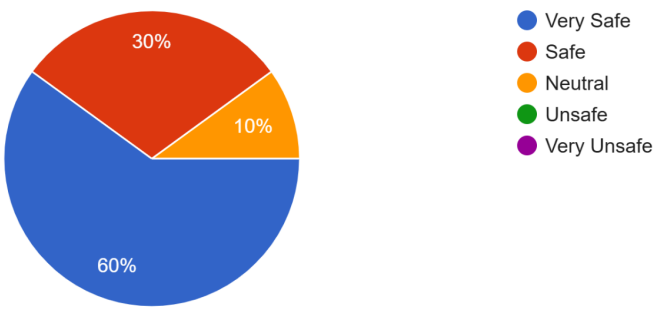


Figure 8: Perceived Safety of the Automated Garage Door System

The survey responses (shown in figure 8) indicate that the majority of users felt secure while using the automated garage door system. When asked, *"How safe did you feel using the automated garage door system?"* 60% of respondents reported feeling "Very Safe," while 30% felt "Safe," highlighting the system's effectiveness in ensuring safety as one of its core features. However, 10% of respondents remained neutral, suggesting that there may still be room for improvements to further enhance user confidence. Overall, 90% of users felt a positive level of safety when operating the system, aligning well with the project's objective to prioritize safety for all users.

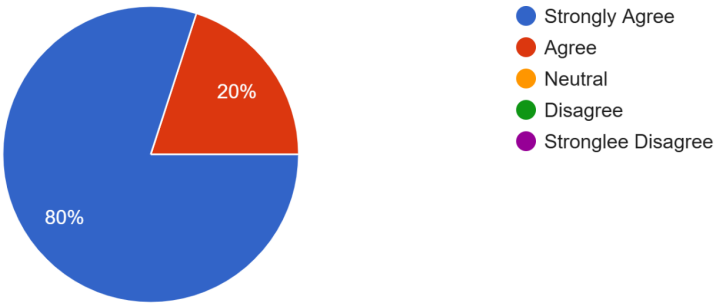


Figure 9: System Accessibility for People with Mobility Challenges

When asked, *"Do you think the system improves accessibility for people with mobility challenges?"* survey results (shown in figure 9) revealed strong support for the system's role in enhancing accessibility for individuals with mobility challenges. An overwhelming 80% of respondents "Strongly Agree" that the automated garage door system significantly improves accessibility, while the remaining 20% "Agree." The absence of neutral or negative responses indicates unanimous approval of the system's effectiveness in addressing accessibility issues. These findings validate the project's objective of creating a hands-free and user-friendly design, particularly benefiting users with physical limitations, thereby demonstrating the system's potential for broader applicability in similar scenarios.

Gantt Chart

	October				November				December			
	5	12	19	26	2-5	13-16	18-23	25-29	1-4	6-7	9-14	16-18
Project Proposal												
Planning												
Designing												
Coding												
Testing/Debugging												
Documentation												
Implementation												

Table 1. Gantt Chart

The Gantt chart (shown in table 1) outlines the project timeline, highlighting key phases such as Proposal, Planning, Designing, Coding, Testing/Debugging, Documentation, and Implementation.

The project starts with defining goals in the Proposal phase, followed by Planning, which involves resource allocation and scheduling. During Designing, the system's architecture is developed, followed by Coding, where the software is built and tested. The Testing/Debugging phase ensures the system functions properly, followed by the Documentation phase for compiling technical details. Finally, the Implementation phase focuses on deploying the completed system. The Gantt chart ensures all phases are completed on schedule and in sequence.

Bibliography

Baichtal, J. (2014). *Arduino for Beginners: Essential Skills Every Maker Needs*. Pearson Education, Inc. , pp. 259-260.

Gaulton, J. (2024, October 21). History of automatic doors | Let's take a look back. *ADC Automatic Doors & Aluminium Windows*. https://adcentrances.com/the-history-of-automatic-doors/

Hasibuan, A., Rosdiana, R., & Tambunan, D. S. (2021). Design and development of an automatic door gate based on internet of things using Arduino Uno. *Neliti*. https://doi.org/10.25008/bcsee.v2i1.1141

Mtshali, P., & Khubisa, F. (2019). *A smart home appliance control system for physically disabled people*. https://www.semanticscholar.org/paper/A-Smart-Home-Appliance-Control-System-for-Disabled-Mtshali-Khubisa/ed86e916216fb7bbfe35b275ad22c5cab8821698

Venkataramanan, V., Shah, D., Panda, I., Shah, S., Davawala, R., Shah, K., & Salot, K. (2023). Smart automatic COVID door opening system with contactless temperature sensing. *e-Prime - Advances in Electrical Engineering Electronics and Energy*, *6*, 100284. https://doi.org/10.1016/j.prime.2023.100284